THE OLD MUTUAL SOUTH AFRICAN MATHEMATICS OLYMPIAD

Organised by the
SOUTH AFRICAN MATHEMATICS FOUNDATION

2024 SECOND ROUND
SENIOR SECTION: GRADE 10 - 12

9 May 2024 **Time: 120 minutes** **Number of questions: 25**

Instructions

1. *The answers to all questions are integers from 000 to 999. Each question has only one correct answer.*
2. *Scoring rules:*
   2.1. *Each correct answer is worth 3 marks in Part A, 4 marks in Part B and 6 marks in Part C.*
   2.2. *There is no penalty for an incorrect answer or any unanswered question.*
3. *You must use an HB pencil. Rough work paper, a ruler and an eraser are permitted.* ***Calculators and geometry instruments are not permitted.***
4. *Figures are not necessarily drawn to scale.*
5. *Indicate your answers on the sheet provided.*
6. *Start when the invigilator tells you to do so.*
7. *Answers and solutions will be available at* `www.samf.ac.za`

Do not turn the page until you are told to do so.
Draai die boekie om vir die Afrikaanse vraestel.

PRIVATE BAG X173, PRETORIA, 0001
TEL: (012) 392-9372 Email: info@samf.ac.za

Organisations involved: AMESA, SA Mathematical Society,
SA Akademie vir Wetenskap en Kuns

The Mathematics Talent Search is a free online problem-solving course for learners from Gr 7 – 12 presented by the SAMF. All you have to do to participate is to click on https://mytutor.chat/samf-talent-search/ or to scan the QR code to go to the MyTutor.chat site easily.

SECTION A (3 MARKS PER QUESTION)

1) Given:

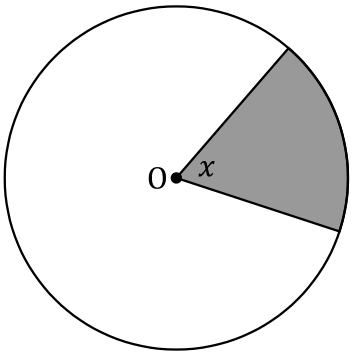
$$3a = 55 - 2b$$
$$10 + 2a = b$$

What is the value of $a + b$?

2) What is the numerator of the largest fraction below?

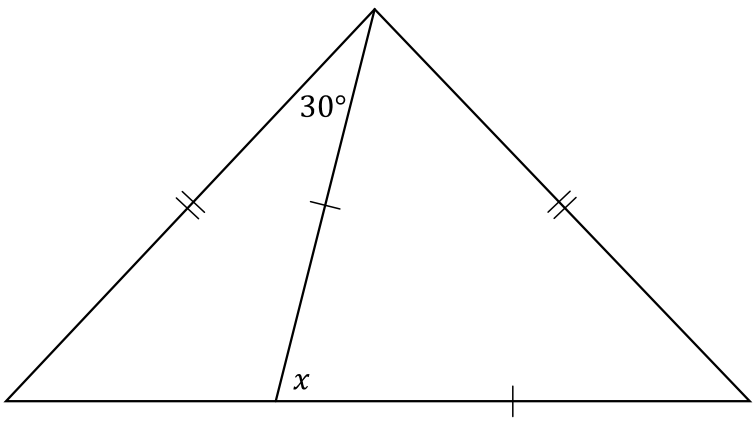
$$\frac{500}{1000} \qquad \frac{501}{1001} \qquad \frac{502}{1002} \qquad \frac{503}{1003} \qquad \frac{504}{1004}$$

3) The diagram shows a circle with center O. If the area of the shaded region is 15% of the area of the circle, what is the value of x in degrees?



4) What is the smallest possible value for $\frac{a}{b} + \frac{b}{a}$ if a and b are positive integers?

5) Calculate angle x, in degrees, in the figure below.



6) What is the sum of the digits if $2^{2024} \cdot 5^{2026}$ is calculated?

7) Calculate $\sqrt[3]{\left(25 + \sqrt[4]{14 + \sqrt{2 + \sqrt[3]{8}}}\right)^4}$.

8) 24 Martians, numbered 1 to 24, stand in a circle in clockwise numerical order, all facing the centre. They begin calling numbers out loud, in clockwise order. The first Martian calls out the number one, the second calls out the number two, and each Martian thereafter calls out the number one more than the previous number which has been called. What is the number of the Martian who calls out the number 2024?

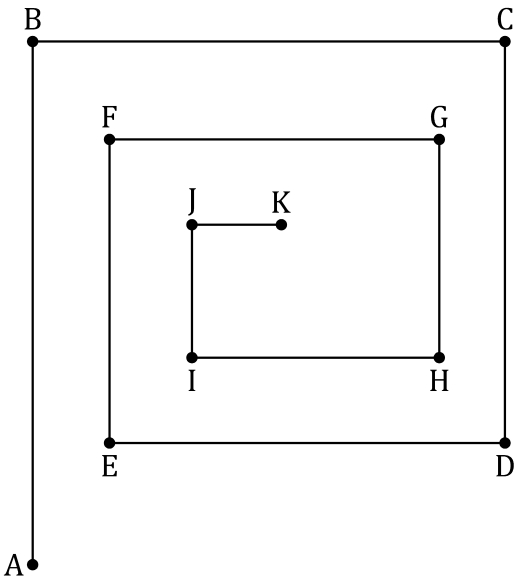
9) What is the largest value of x such that 2^x divides exactly into

$$19 \times 18 \times 17 \times \cdots \times 3 \times 2 \times 1?$$

10) 10 people stand in a row. The first person goes to the back of the row and the next person leaves the row, so that the person who was third in the row is now first in the row. Now the person first in the row goes to the back of the row, and the next person leaves the row. This process is repeated until only one person remains. What was the original position in the row of the only remaining person?

SECTION B (4 MARKS PER QUESTION)

11) In the set of perpendicular line segments shown on the right, $AB = 10$ and every other segment is 1 unit shorter than the previous one. So $BC = 9$, $CD = 8$, $DE = 7$, etc. What is the square of the length of line segment AK?



12) In the table, the product of the positive integers, represented by the letters in each row and column, are given. For example, $xy = 9$ and $xz = 45$. What is the value of xw?

x	y	9
z	w	80
45	16	

13) The Mug Shop keeps all kinds of different, interesting mugs. The shopkeeper starts working at 08:00 and stops at 13:00. A certain collection contains 6 mugs put next to each other on a shelf. He takes 55 minutes to arrange all the (remaining) mugs of this collection in all possible different arrangements. If he starts arranging the mugs from this collection exactly on the hour every hour and one mug from this collection gets sold in the last five minutes of each hour, how many different arrangements did the shopkeeper make in one day?

14) How many nines are there in 99999^2 after it has been calculated?

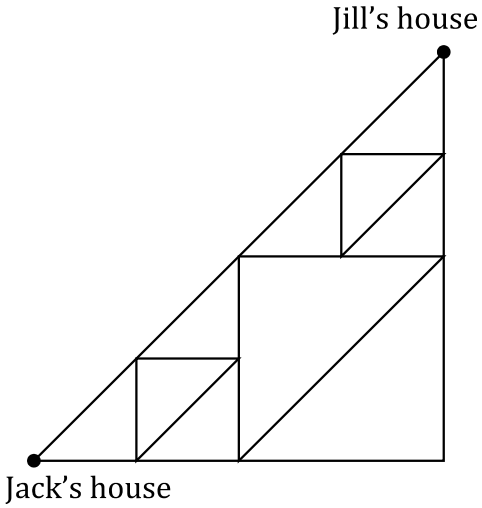
15) A train, moving at a constant speed, takes 4 seconds to cross the equator. It completely passes a platform, 75 m long, in 9 seconds. What is the length of the train in metres?

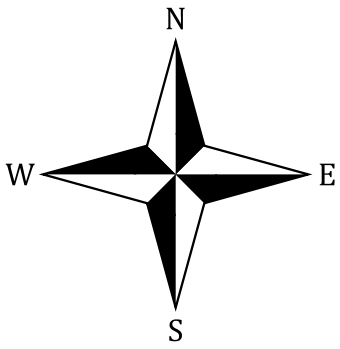
16) Let x and y be real numbers in the equations

$$x^3 + 3xy^2 = 28, \text{ and}$$
$$y^3 + 3yx^2 = 26.$$

What is the value of $x^6 + y^6$?

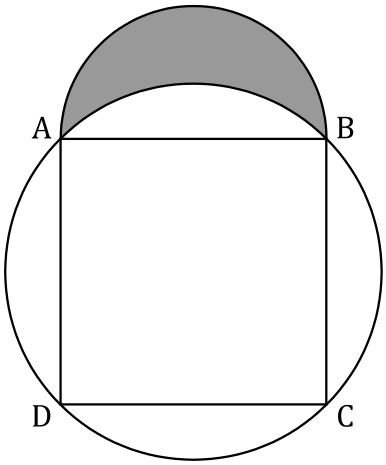
17) Jill and Jack's houses are at the opposite ends of a park. Jack bikes to Jill's house along the paths laid out in the park, as shown in the diagram. In how many different ways can Jack get to Jill's house if he can only move North, East or North-East?





18) If $f(1) = 1$ and $f(n) = n + f(n-1)$ for all natural numbers $n \geq 2$, determine $f(30)$.

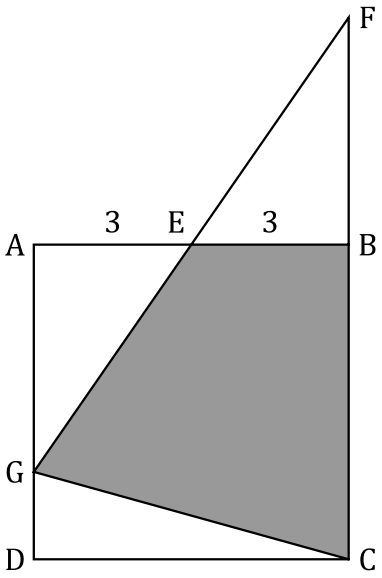
19) The circumcircle of square $ABCD$ has radius 10. A semicircle is drawn on AB, outside the square. Determine the area of the shaded region.



20) Burgers are sold plain or with any selection of the available toppings. Burgers with a different selection of toppings (including with none) are considered different types. If Rocoburger has 192 more types of burgers than Burgerbraai, but uses only 2 more toppings than Burgerbraai, how many types of burgers does Burgerbraai have?

SECTION C (6 MARKS PER QUESTION)

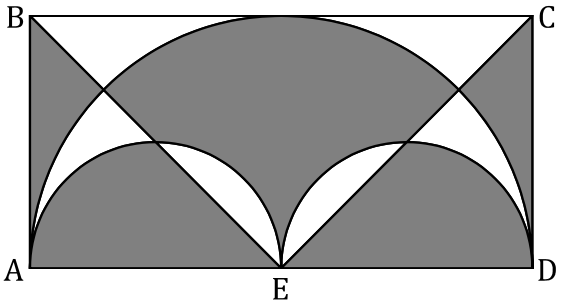
21) In square $ABCD$ in the figure on the right, point G lies on AD and point E lies on AB such that $AE = BE = 3$. The line segment GE extended meets CB extended in point F such that $E\hat{G}C = B\hat{C}G$. What is the area of the shaded region in the figure?



22) Consider the sequence defined by $a_k = \frac{1}{k^2+k}$ for $k \geq 1$. Given that $a_m + a_{m+1} + \cdots + a_{n-1} = \frac{1}{29}$ for positive integers m and n with $m < n$, find $m + n$.

23) In trapezium $ABCD$ with AB parallel to CD, the diagonals intersect at P. The area of $\triangle ABP$ is 72 cm^2 and the area of $\triangle CDP$ is 50 cm^2. Find the area of the trapezium, in cm^2.

24) $ABCD$ in the figure is a rectangle with inscribed semicircle on AD. Semicircles on AE and DE touch each other in point E on AD, and rectangle $ABCD$ respectively in point A and point D. Both BE and CE are straight line segments. If $CD = 2$ and $AE = DE$, calculate the total area of all shaded regions.



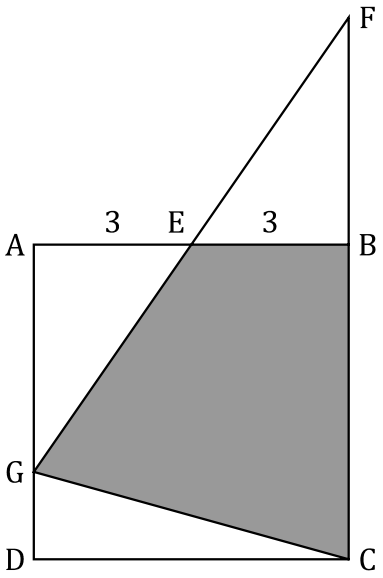
25) Two different positive integers a and b satisfy the equation

$$a^2 - b^2 = 2024 - 2a$$

How many pairs (a, b) of numbers satisfy the equation?

AFDELING C (6 PUNTE PER VRAAG)

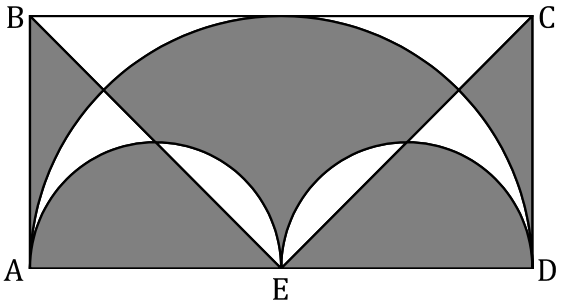
21) In vierkant $ABCD$ in die figuur hiernaas, lê punt G op AD en punt E op AB sodanig dat $AE = BE = 3$. Die lynsegment GE verleng ontmoet CB verleng in punt F sodanig dat $E\hat{G}C = B\hat{C}G$. Wat is die oppervlakte van die ingekleurde deel in die figuur?



22) Beskou die ry wat gedefinieer word deur $a_k = \frac{1}{k^2+k}$ vir $k \geq 1$. Gegee dat $a_m + a_{m+1} + \cdots + a_{n-1} = \frac{1}{29}$ vir positiewe heelgetalle m en n waar $m < n$, bepaal $m + n$.

23) In trapesium $ABCD$ met AB ewewydig aan CD, sny die hoeklyne in P. Die oppervlakte van $\triangle ABP$ is 72 cm² en die oppervlakte van $\triangle CDP$ is 50 cm². Bereken die oppervlakte van die trapesium, in cm².

24) $ABCD$ in die figuur is 'n reghoek met 'n ingeskrewe semisirkel op AD. Semisirkels AE en DE raak mekaar by punt E op AD en reghoek $ABCD$ onderskeidelik by punt A en punt D. Beide BE en CE is reguitlyne. As $CD = 2$ en $AE = DE$, bereken die totale oppervlakte van al die ingekleurde dele tesame.

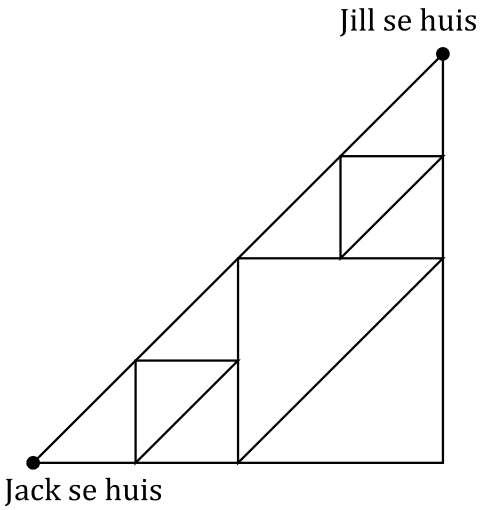


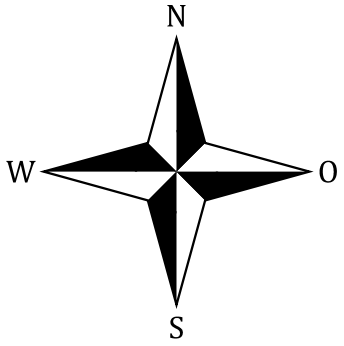
25) Twee verskillende positiewe heelgetalle a en b bevredig die vergelyking

$$a^2 - b^2 = 2024 - 2a$$

Hoeveel pare getalle (a, b) bevredig die vergelyking?

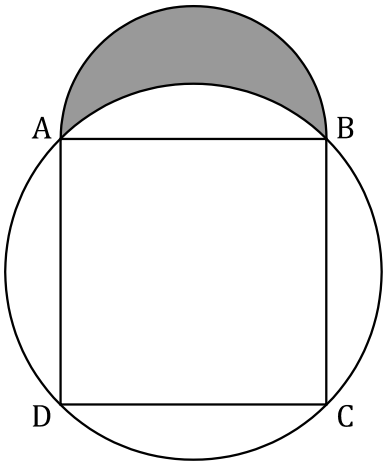
17) Jill en Jack se huise is aan die teenoorgestelde punte van 'n park geleë. Jack ry per fiets na Jill se huis op die paaie wat in die park uitgelê is, soos in die diagram aangedui. Op hoeveel verskillende maniere kan Jack na Jill se huis toe ry as hy slegs Noord, Oos of Noordoos kan beweeg?





18) Indien $f(1) = 1$ en $f(n) = n + f(n-1)$ vir alle natuurlike getalle $n \geq 2$, bepaal $f(30)$.

19) Die omgeskrewe sirkel van vierkant $ABCD$ se radius is 10. 'n Semisirkel word buite die vierkant op AB geteken. Bepaal die oppervlakte van die ingekleurde deel.



20) Burgers word met geen of met enige keuse van beskikbare vulsels verkoop. Burgers met verskillende vulsels (ingesluit geen) word as verskillend beskou. As Rocoburger 192 variasies burgers meer as Burgerbraai het, maar net 2 meer vulsels as Burgerbraai, hoeveel variasies van burgers het Burgerbraai?

12) In die tabel word die produk van die positiewe heelgetalle, wat deur die letters in elke ry en kolom voorgestel word, gegee. Byvoorbeeld, $xy = 9$ en $xz = 45$. Wat is die waarde van xw?

x	y	9
z	w	80
45	16	

13) Die Mug Shop hou allerlei verskillende, interessante bekers aan. Die winkelier begin om 08:00 werk en stop om 13:00. 'n Sekere versameling bevat 6 bekers wat langs mekaar op 'n rak uitgepak is. Hy neem 55 minute om al die (oorblywende) bekers van hierdie versameling in alle moontlike rangskikkings te plaas. As hy elke uur presies op die uur begin om die bekers van hierdie versameling te rangskik en een beker van hierdie versameling verkoop gedurende die laaste vyf minute van elke uur, hoeveel verskillende rangskikkings het die winkelier in een dag gemaak?

14) Hoeveel neges is daar in 99999^2 nadat dit uitgewerk is?

15) 'n Trein wat teen 'n konstante spoed beweeg, ry oor die ewenaar in 4 sekondes. In 9 sekondes beweeg dit ten volle verby 'n platvorm wat 75 m lank is. Wat is die lengte van die trein in meter?

16) Laat x en y reële getalle in die vergelykings

$$x^3 + 3xy^2 = 28, \text{ en}$$
$$y^3 + 3yx^2 = 26$$

wees. Wat is die waarde van $x^6 + y^6$?

6) Wat is die som van die syfers van $2^{2024} \cdot 5^{2026}$ wanneer dit uitgewerk word?

7) Bereken $\sqrt[3]{\left(25 + \sqrt[4]{14 + \sqrt{2 + \sqrt[3]{8}}}\right)^4}$.

8) 24 Marsmannetjies, genommer van 1 tot 24, staan in 'n sirkel in kloksgewyse numeriese volgorde en almal kyk na die middel. Hulle begin getalle uitroep, kloksgewys in numeriese volgorde. Die eerste Marsmannetjie roep die getal een uit, die tweede die getal twee en elkeen daarna roep die getal uit wat een meer is as die vorige getal wat uitgeroep is. Wat is die nommer van die Marsmannetjie wat die getal 2024 uitroep?

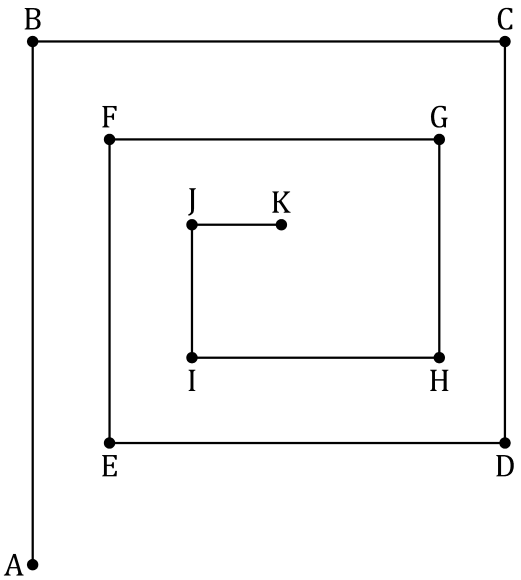
9) Wat is die grootste waarde van x sodanig dat 2^x presies indeel in

$$19 \times 18 \times 17 \times \cdots \times 3 \times 2 \times 1?$$

10) 10 mense staan agter mekaar in 'n ry. Die voorste persoon beweeg na agter in die ry en die volgende persoon verlaat die ry, sodat die persoon wat derde in die ry was nou voor is. Die persoon wat nou voor in die ry is, beweeg na agter in die ry en die volgende persoon verlaat die ry. Hierdie proses word herhaal totdat slegs een persoon oorbly. Wat was die oorspronklike posisie van die oorblywende persoon in die ry?

AFDELING B (4 PUNTE PER VRAAG)

11) In die stel loodregte lynsegmente hiernaas vertoon, is $AB = 10$ en elke ander segment is 1 eenheid korter as die vorige een. So $BC = 9$, $CD = 8$, $DE = 7$, ens. Wat is die kwadraat van die lengte van die lynsegment AK?



AFDELING A (3 PUNTE PER VRAAG)

1) Gegee:

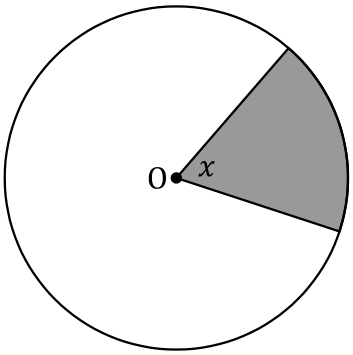
$$3a = 55 - 2b$$
$$10 + 2a = b$$

Wat is die waarde van $a + b$?

2) Wat is die teller van die grootste breuk hieronder?

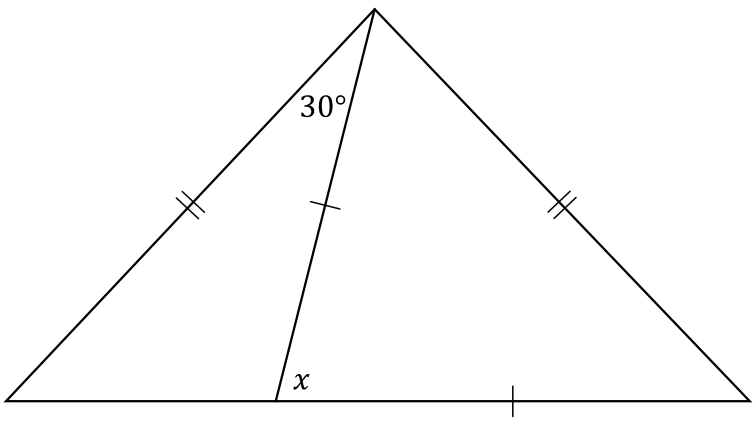
$$\frac{500}{1000} \qquad \frac{501}{1001} \qquad \frac{502}{1002} \qquad \frac{503}{1003} \qquad \frac{504}{1004}$$

3) Die diagram vertoon 'n sirkel met middelpunt O. As die oppervlakte van die ingekleurde deel 15% van die oppervlakte van die sirkel is, wat is die waarde van x in grade?



4) Wat is die kleinste moontlike waarde van $\frac{a}{b} + \frac{b}{a}$ as a en b beide positiewe heelgetalle is?

5) Bereken hoek x, in grade, in die figuur hieronder.



DIE OLD MUTUAL SUID-AFRIKAANSE WISKUNDE-OLIMPIADE

Georganiseer deur die
SOUTH AFRICAN MATHEMATICS FOUNDATION

2024 TWEEDE RONDTE
SENIOR AFDELING: GRAAD 10-12

9 Mei 2024 **Tyd: 120 minute** **Aantal vrae: 25**

Instruksies

1. *Die antwoorde op al die vrae is heelgetalle van 000 tot 999. Elke vraag het slegs een korrekte antwoord.*
2. *Puntetoekenning:*
   2.1. *Elke korrekte antwoord tel 3 punte in Afdeling A, 4 punte in Afdeling B en 6 punte in Afdeling C.*
   2.2. *Geen punte word afgetrek vir foutiewe antwoorde of onbeantwoorde vrae nie.*
3. *Gebruik 'n HB potlood. Papier vir rofwerk, 'n liniaal en uitveër word toegelaat.* ***Sakrekenaars en meetkunde-instrumente word nie toegelaat nie.***
4. *Figure is nie noodwendig volgens skaal geteken nie.*
5. *Beantwoord die vrae op die antwoordblad wat voorsien word.*
6. *Begin sodra die toesighouer die teken gee.*
7. *Antwoorde en oplossings sal beskikbaar wees by* `www.samf.ac.za`

Moenie omblaai voordat dit aan jou gesê word nie.
Turn the booklet over for the English paper.

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Organisasies betrokke: AMESA, SA Wiskundevereniging,
SA Akademie vir Wetenskap en Kuns

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